

ISABELLA VERA

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EDUCATION

Harvey Mudd College, Claremont, CA

Expected May 2028

B.S. - General Engineering, GPA: 3.466, Dean's List

Relevant Coursework: Digital Electronics and Computer Engineering, Engineering Design and Manufacturing, Experimental Engineering, Electronic and Magnetic Circuits and Devices

Awards & Recognition: President's Scholars Program Award Recipient, Annenberg Leadership Fellowship

PROJECTS

Embedded Real-Time Control System for Multi-Channel Linear Actuator Array

February 2026 - Present

- Designing and implementing a microcontroller-based control system to convert real-time mapping sensor data into synchronized multi-channel actuator commands
- Testing and iterating on system performance to reduce latency and improve spatial resolution

Autonomous Underwater Vehicle

January 2026 - Present

- Designing and implementing custom temperature, pressure, and turbidity sensor circuits on an AUV to facilitate ocean conservation data collection
- Programming AUV to traverse open water to collect data without human intervention, and return to launch point

RISC-V Multi-cycle Processor

November 2025 - December 2025

- Built and debugged a multi-cycle digital system with centralized control, shared memory, and staged execution, emphasizing system reliability and correct sequencing of operations using SystemVerilog
- Verified correct system behavior through waveform-based debugging and testbench-driven validation, diagnosing timing, control, and data-path failures

Logic Gate Demonstration Board

March 2025 - May 2025

- Headed a team of 5 students in designing a demonstration board to teach functions of basic logic gates
- Fabricated a functioning prototype with knowledge of laser cutters, 3D printers, and integrated circuits to help over a hundred students each semester in understanding material

Harvey Mudd Engineering Hammer

March 2025 - April 2025

- Machined a hammer from raw stock using lathe, mill, band saw, and sander, to tight specifications and tolerances

Code the Change Website Team

December 2024 - February 2025

- Collaborated with a team of 4 other students to launch a website for Boston's Museum of Science and MIT's Jameel clinic to educate global audiences on clinical artificial intelligence

PROFESSIONAL EXPERIENCE

Electropenetography Firmware Developer, USDA

March 2026 - Present

- Develop microcontroller firmware and Python-based software to acquire, process, and visualize high-sensitivity analog signals from insect feeding experiments

Engineering Grader and Tutor, Harvey Mudd College Engineering Dept.

January 2026 - Present

- Conduct weekly office hours to assist students in completing homework and hands-on lab assignments

Steward, Harvey Mudd College Makerspace

January 2025 - Present

- Lead biweekly trainings reviewing operation and safety procedures of waterjet cutters
- Manage inventory and organization of electronics area to ensure all students a low barrier to entry to high level engineering equipment
- Leverage Google Apps Script to create a budget analysis program to manage a six figure yearly budget

Lead STEM Tutor, Partnership for Success

June 2024 - August 2024

- Taught an engineering class for 30 girls from underprivileged communities to introduce to design thinking and STEM

SKILLS

Collaboration, Leadership, Project Management, Communication, Organization, Microsoft Office

Design & Manufacturing: SolidWorks, 3D Printing, Waterjet, Lathe, Mill, Soldering, Electronics Test Equipment

Programming: JavaScript, HTML, CSS, Python, Java, C#, Racket, C, SystemVerilog